Hydraulic Power Package (Mod 6/2039) – General Data

1. General

The hydraulic power package supplies hydraulic pressure to operate the wing flaps, wheel brakes, nose wheel steering systems. Refer to Figure 1 for details.

2. Description

The hydraulic power package consists of a tray containing the reservoir, brake and airframe damping accumulators, indicators and charging valves, electric motor-driven hydraulic pump with thermal relief switch, system filter, system pressure relief valve, brake pressure relief valve, system pressure switch, and associated hydraulic lines and check valves. The package is mounted on the fuselage structure beneath the flight compartment floor and is accessible from both sides of the fuselage through access doors.

A. Hydraulic Reservoir

The hydraulic reservoir is a cylindrical tank containing a filler neck, a filter screen, a cap and dipstick assembly, two suction outlets, a return inlet, and a vent. The reservoir is bolted to the power package tray. The filler neck, filter screen, and cap and dipstick assembly are accessible through the left access opening.

B. Airframe System Accumulator

The airframe system accumulator is incorporated in the hydraulic system to dampen out pressure surges in the hydraulic system and provide a secondary source of operating pressure. The accumulator is secured to the power package tray by two clamps. A combined indicator and air charging valve for the airframe system accumulator is mounted on a bracket on the left side of the power package.

C. Wheel Brakes Accumulator

The wheel brakes accumulator provides the wheel brakes with supplemental pressure in an emergency or during limited periods of peak demand, and provides pressure when the hydraulic system is shutdown while parked. The accumulator is secured to the power package tray by two clamps. A combined indicator and air charging valve for the wheel brakes accumulator is mounted on a bracket on the left side of the power package.

D. Electric Motor-Driven Hydraulic Pump

The electric motor-driven hydraulic pump is mounted on the hydraulic power package tray, and is the primary source of power for the hydraulic system. The pump consists of a motor coupled gear type pump. The motor operates on 28 Volt DC power, and is supplied from the left DC bus. A 35-ampere circuit breaker protects the circuit. The motor is equipped with a radio noise suppressor, and a thermal relief switch.

E. Hydraulic System Filter

The hydraulic system filter is a tee type containing a replaceable 10-micron element, and is installed in the hydraulic power package. Provision is made to bypass the element should a blockage or restriction cause a pressure differential of 50 ± 5 psi maximum between inlet and outlet.

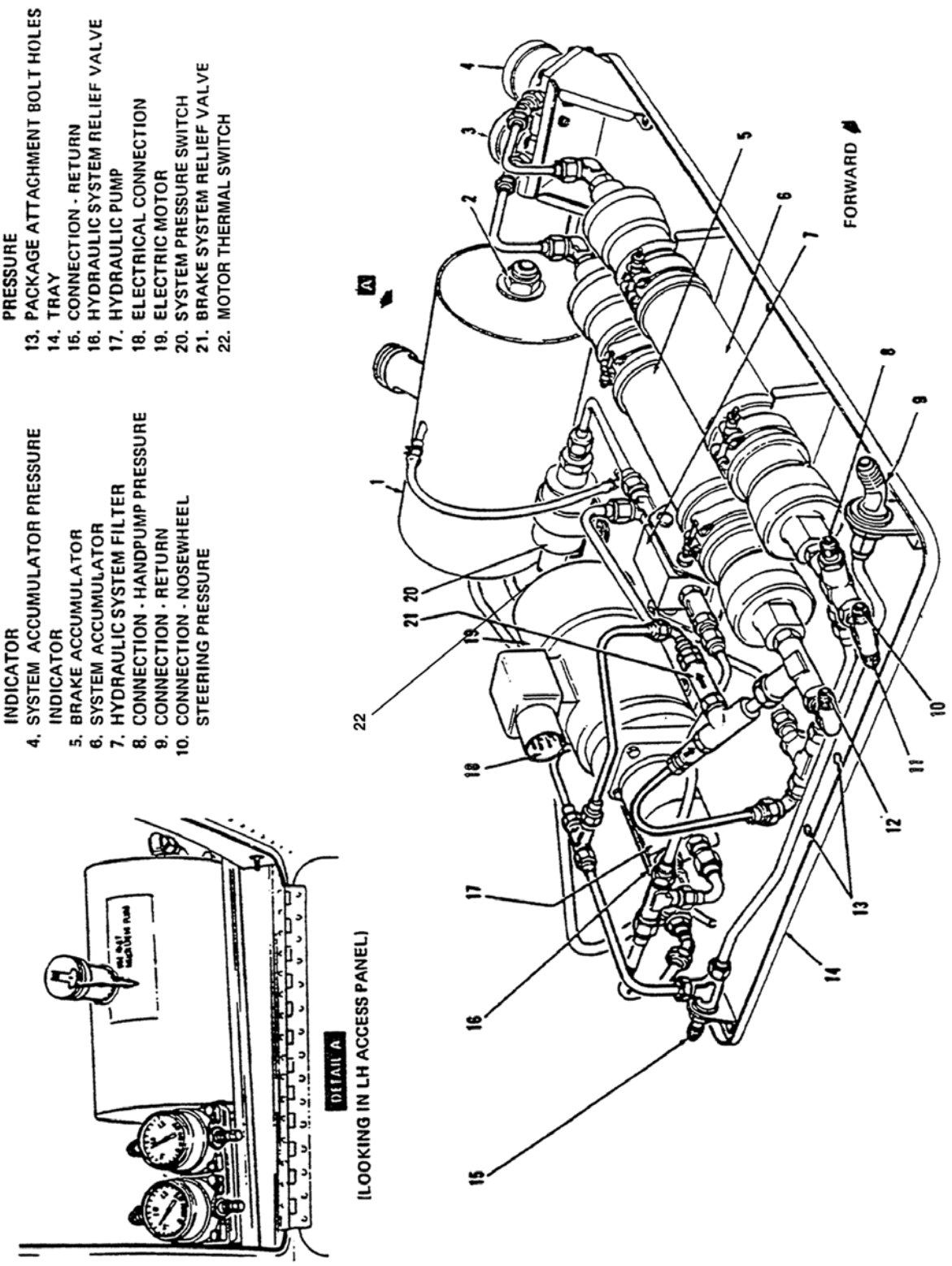
F. Pressure Switch

The pressure switch (Pre Mod 6/2313), mounted on the hydraulic power package tray, electrically controls the motor-driven hydraulic pump. It regulates the working pressure of the hydraulic system by switching off DC power to the motor when the system pressure reaches 1575 ± 50 psi and switches on DC power when the pressure drops to 1225 psi minimum.

The pressure switch (Post Mod 6/2313 or Post Mod 6/2456), mounted on the hydraulic power package tray, electrically controls the motor-driven hydraulic pump. It regulates the working pressure of the hydraulic system by switching off DC power to the motor when the system pressure reaches approximately 1575 psi and switches on DC power when the pressure drops to approximately 1225 psi minimum.

G. Pressure Relief Valve

Two pressure relief valves are installed, one for airframe hydraulic system, and the other for brake system. Both valves are located on the hydraulic power package, and are set to relieve excessive pressure buildup. The airframe hydraulic system relief valve relieves pressure buildup in excess of 1950 psi at full flow. The brake system relief valve is set to crack at a pressure of 1750 psi.



Hydraulic Power Package
Figure 1

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3. Removal/Installation

Refer to Figure 2 for details.

A. Remove Hydraulic Power Package

1) Discharge airframe system accumulator hydraulic pressure by operating wing flaps, and wheel brakes accumulator hydraulic pressure by pumping brake pedals.

2) Open left and right-hand access panels on underside of nose section (refer to 12-00-00, Servicing – General Data).

3) Ensure electric power supply is OFF. Disconnect electrical connectors at hydraulic pump motor and pressure switch.

4) Discharge all air pressure from airframe system accumulator and wheel brakes accumulator by loosening each air valve approximately one-quarter turn, and wait until all air has escaped. Check indicators of both accumulators read zero.

WARNING FAILURE TO PROPERLY DISCHARGE ALL AIR PRESSURE FROM AN ACCUMULATOR MAY RESULT IN INJURY TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

5) Place container to receive spillage. Disconnect return lines at elbow (on forward bracket) and tee (on aft bracket) on right-hand side of hydraulic power package tray.

6) Disconnect pressure lines to nose wheel steering, wing flaps, hand pump, and wheel brakes.

7) Disconnect hand pump suction line at reservoir.

8) Remove seven screws and washers attaching tray to structure, and remove power package through left-hand access panel.

9) Cap all open lines or fittings as required.

B. Install Hydraulic Power Package

1) Position power package and secure tray to aircraft structure with seven screws and washers.

2) Connect handpump suction line at reservoir.

3) Connect wheel brake, handpump, wing flaps, and nose wheel steering pressure lines.

4) Connect return lines at elbow (on forward bracket) and tee (on aft bracket) on right-hand side of power package tray.

5) Connect hydraulic pump motor and pressure switch electrical connectors.

6) Refill reservoir with hydraulic fluid, Specification MIL–H–5606.

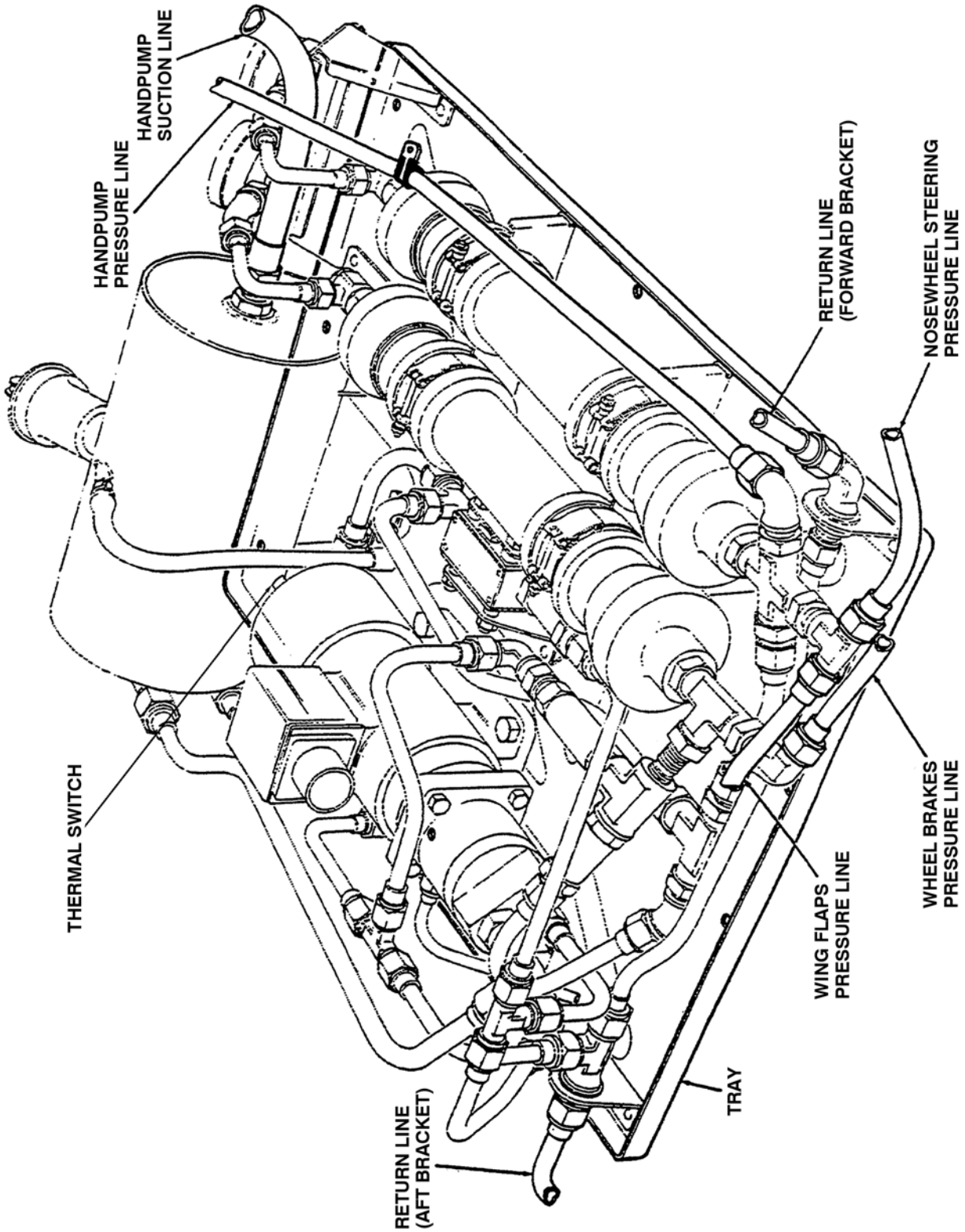
7) Recharge airframe system accumulator and wheel brake accumulator with dry nitrogen to 750 ± 50 psi.

8) Carry out function test of hydraulic power package (refer to Para 4.). Bleed handpump (refer to 29-20-11, Hydraulic Handpump (Mod 6/2093) – Maintenance Practices).

9) Bleed nose wheel steering, wing flaps, and wheel brakes systems (refer to 27-50-11, Flap Actuator – Maintenance Practices, 32-40-00, Wheels and Brakes (Mod 6/2002) – General Data and 32-50-00, Nose Wheel Steering System – General Data).

10) Check all joints in hydraulic power package for leaks. Retighten as necessary.

11) Close left and right-hand access panels (refer to 12-00-00, Servicing – General Data).



Hydraulic Power Package Installation
Figure 2

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4. Adjustment/Test

A. Function Test Hydraulic Power Package (Pre Mod 6/2313)

1) Open left and right-hand access panels on underside of nose section (refer to 12-00-00, Servicing – General Data).

2) Discharge airframe hydraulic system pressure by operating wing flaps, and brake system hydraulic pressure by pumping brake pedals. Check indicators on both accumulators read 750 ± 50 psi. Recharge accumulators with dry nitrogen if necessary.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Check nose wheel steering lever corresponds with position of nose wheel, wing flaps selector with position of wing flaps, and set parking brake handle to park.

4) Check that electric motor-driven hydraulic pump charges airframe system accumulator and wheel brakes accumulator, and that motor cuts out when pressure reading on indicator is 1575 ± 50 psi.

5) Operate wing flaps. Check that motor cuts in when reading on airframe hydraulic system pressure indicator drops to 150 – 300 psi below initial setting.

6) Pump brake pedals. Check that motor cuts in when reading on brake system pressure indicator drops to 150 – 300 psi below initial setting. Reset parking brake handle to park.

7) Check all joints on hydraulic power package for leaks. Retighten connections if necessary.

8) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

9) Close left and right-hand access panels (refer to 12-00-00, Servicing – General Data).

B. Function Test Hydraulic Power Package (Post Mod 6/2313) (Post Mod 6/2456)

1) Open left and right-hand access panels on underside of nose section (refer to 12-00-00, Servicing – General Data).

2) Discharge airframe hydraulic system pressure by operating wing flaps, and brake system hydraulic pressure by pumping brake pedals. Check indicators on both accumulators read 750 ± 50 psi. Recharge accumulators with dry nitrogen if necessary.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Check nose wheel steering lever corresponds with position of nose wheel, wing flaps selector with position of wing flaps, and set parking brake handle to park.

4) Check that electric motor-driven hydraulic pump charges airframe system accumulator and wheel brakes accumulator, and that motor cuts out when pressure reading on indicator reaches <1700 psi.

5) Operate wing flaps. Check that motor cuts in when reading on airframe hydraulic system pressure indicator drops to 150 – 350 psi below initial setting and >1225 psi.

6) Pump brake pedals. Check that motor cuts in when reading on brake system pressure indicator drops to 150 – 350 psi below initial setting and >1225 psi. Reset parking brake handle to park.

7) Check all joints on hydraulic power package for leaks. Retighten connections if necessary.

8) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

9) Close left and right-hand access panels (refer to 12-00-00, Servicing – General Data).

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